

Literature Status for the Two-Dimensional q -Araki-Woods Factoriality Gap in arXiv:1703.01317

Run fa_banach_001, lane 14

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Status

This note records that the factoriality and non-injectivity gap for two-dimensional q -Araki-Woods algebras flagged in Avsec–Brannan–Wasilewski, arXiv:1703.01317, is answered by Kumar–Skalski–Wasilewski, arXiv:2301.08619.

No new theorem is claimed here. The purpose of this packet is to clear an already-answered open-problem signal from the run queue.

Source Question

In the introduction of arXiv:1703.01317, page 2, the authors survey known structural results for q -Araki-Woods algebras. They state that for the factoriality problem and non-injectivity, the remaining case is the family of q -Araki-Woods algebras built from a two-dimensional Hilbert space. This is the open-problem signal extracted from source line 140 of the local TeX source.

Later Answer

The abstract of arXiv:2301.08619 says that the paper establishes factoriality of q -Araki-Woods von Neumann algebras with at least two generators in full generality, and also establishes non-injectivity and the type of the factors.

The main theorem on page 2 states that if $q \in (-1, 1)$, $\dim H_{\mathbb{R}} \geq 2$, and $(U_t)_{t \in \mathbb{R}}$ is a group of orthogonal transformations of $H_{\mathbb{R}}$, then $\Gamma_q(H_{\mathbb{R}}, U_t)$ is a non-injective factor, with type determined by the closed subgroup of $\mathbb{R}_*^{\times}$ generated by the spectrum of the generator of $(U_t)_{t \in \mathbb{R}}$. Theorem 2.5 on page 9 records non-injectivity for $\dim H_{\mathbb{R}} \geq 2$.

Therefore the two-dimensional case singled out in arXiv:1703.01317 is already covered by later literature. The answer paper explicitly frames itself as a full solution of the q -Araki-Woods factoriality question, so this packet is classified as `literature_already_answered`.

Scope

Resolved scope: factoriality and non-injectivity for q -Araki-Woods algebras with underlying real Hilbert-space dimension at least two, including the two-dimensional case named in arXiv:1703.01317.

Not resolved by this packet: the neighboring radial-multiplier conjecture in arXiv:1703.01317, namely type-invariance and the expected isometric and bijective form of the transference principle for radial multipliers.

Search Evidence

The lane checked the run indexes for arXiv:1703.01317 and core keywords including q -Araki-Woods, two-dimensional Hilbert space, factoriality, non-injectivity, and radial multipliers. No prior packet for arXiv:1703.01317 was found. External arXiv/web checks identified arXiv:2301.08619 as the decisive later answer. The later mixed q -Araki-Woods paper arXiv:2304.11108 also records that the complete factoriality and non-injectivity question for the non-mixed q -Araki-Woods algebras had been settled by Kumar–Skalski–Wasilewski.

References

1. S. Avsec, M. Brannan, and M. Wasilewski, *Complete metric approximation property for q -Araki-Woods algebras*, arXiv:1703.01317; J. Funct. Anal. 274 (2018), no. 2, 544–572.
2. M. Kumar, A. Skalski, and M. Wasilewski, *Full solution of the factoriality question for q -Araki-Woods von Neumann algebras via conjugate variables*, arXiv:2301.08619; Comm. Math. Phys. 402 (2023), 157–167.
3. M. Kumar, *Conjugate variables approach to mixed q -Araki-Woods algebras: Factoriality and non-injectivity*, arXiv:2304.11108; J. Math. Phys. 64 (2023), 093506.